UoM Student Safety Project

SYSTEM ANALYSIS AND DESIGN 7610 M50

Presented by:

Group 3

Koneru, Srikedhar Bhagavan Sankar
Kumara, Janaki Ram
Lakshminaryana, Aishwarya
Madala, Nikitha
Masini, Vijaya Durga
Mohith Abhay Krishna Kolusu

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INTRODUCTION

Do you think safety is important?

What are the facilities provided by the university to maintain safety and security of individual?

Are you aware of it?

Do you think we can improve?



INTRODUCTION

Availability:

UoM have Blue light phone, Red phones, digital cameras along with 3 safety services like Ever bridge, get home safety and tiger patrol with good patrolling facility.

Improvements:

We have designed a simple system for students and even faculties of the university that provides safety in one click.

Planning Phase

Problem Statement

- Some students face threats, physical and emotional violence, theft, insecurities in and around university environments.
- It gives negative impact on their learnings and so do for their mental health.

Objectives

- To provide safe environment for students and faculties in and around campus.
- To prevent harm or damages and reduce feel of unsafety.
- To provide easy access for securing one another.
- To create awareness among peers.

Planning Phase

Technical feasibility:

- It should be able to host on university's intranet or can be extended to my Memphis portal.
- HTML, CSS, JavaScript for front end and PHP, MySql as backend.
- Integrating with other services like google maps along with encrypted authentication be facilitated.

Economic feasibility:

- Low cost of development since it is extension for existing system.
- Low maintenance cost for storage and security since we are using available resource.
- It is free service offered by the university without thinking about return on investment.

Analysis phase

Drawbacks of existing system:

- Time taken for actions is more
- Viewing Digital camera for 24/7 and then reaching that spot in emergency is difficult.
- Awareness provided to students are not sufficient.

Goal/ Motive:

- Provide reliable platform to report incident.
- Offer guides, tips and contact details for emergency services.
- User friendly interface with improved and maintenance of secure tracking.
- Allow administrators for easy and quick analysis of incident for taking action with strong encryption and safeguarding student’s data and privacy.

Analysis Phase

Proposed system:

- UoM student safety portal is designed working with simple HTML, CSS and JavaScript creating user friendly interface.
- It is reliable reporting system that improves student security in and around campus with faster emergency response.
- It provides stable connectivity among community and increases awareness.

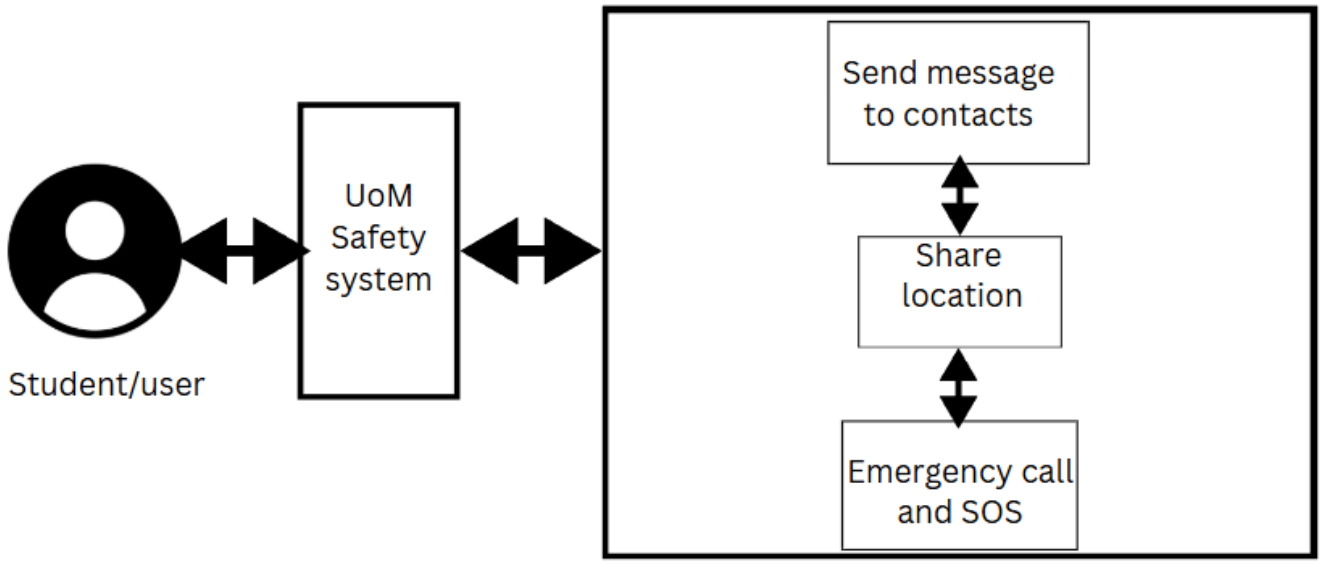
Benefits of Proposed system:

- Offers real time alerts and safe communication directly with authorities possible.
- Acts as centralized platform for students.
- It integrates various safety features, emergency response mechanism.
- It is more efficient and reliable offering quick way to take action

System Architecture:

- User gives input details to system and get acess, First create an account and fill details.
- It have features to send message to contacts, share location, make emergency call SOS and notification.
- There is direct emergency call made for university cops or send SOS in worst case.

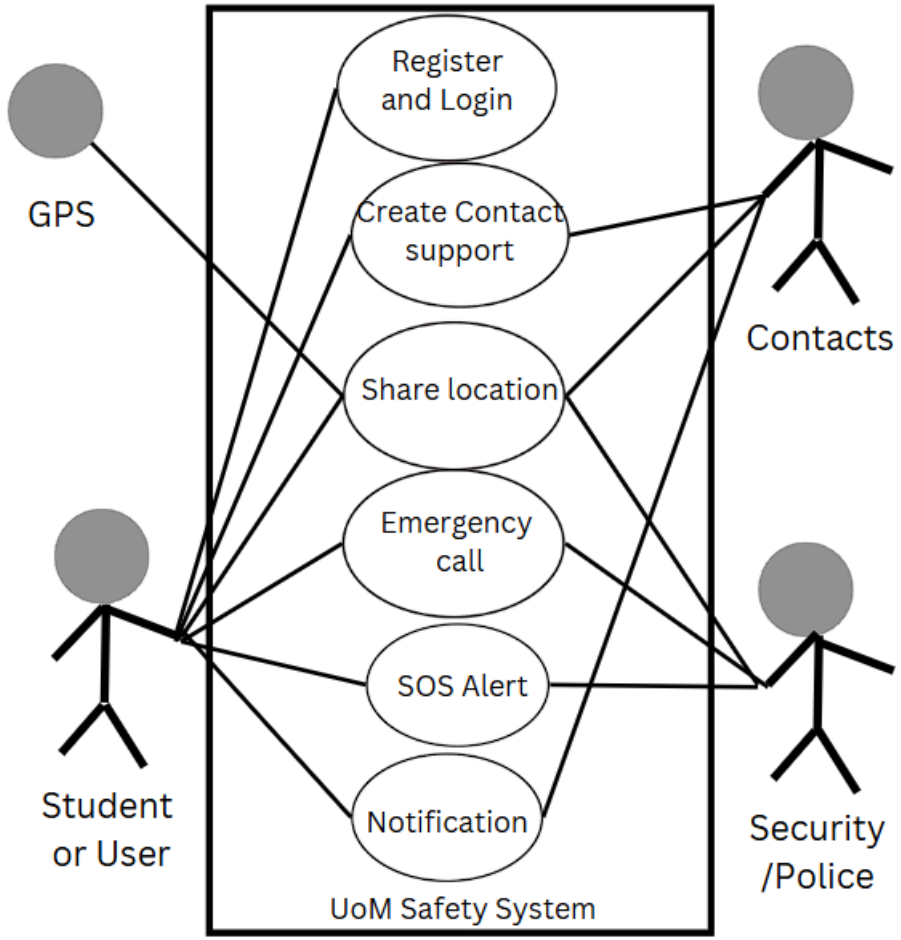
Design Phase



Design Phase

Usecase diagram:

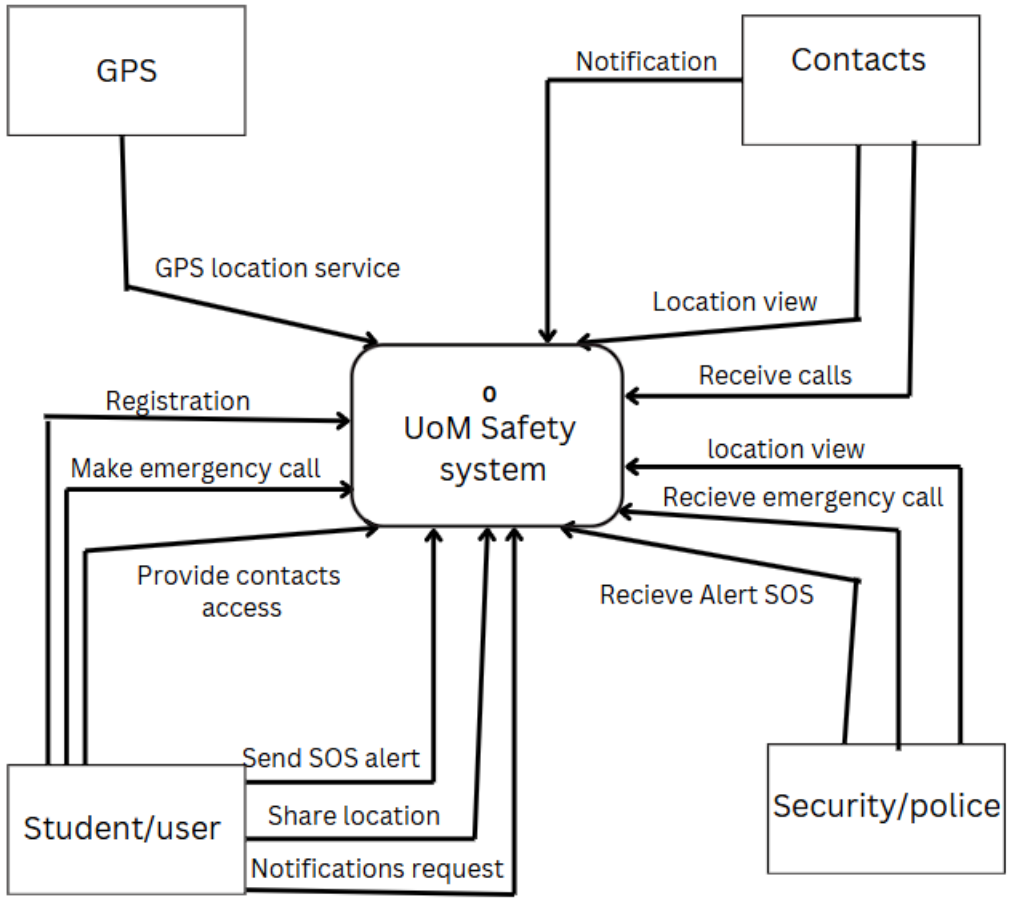
- Actors: Student or primary users access the application.
- GPS provide live location when user is making emergency call or sos or simply needs to check their location.
- Contacts, Security/ Police are people who support user in need and safeguard them.
- Cases: Register/Login, Create contact support, Share location, Emergency call, SOS alert and notification.



Design Phase

Dataflow diagram:

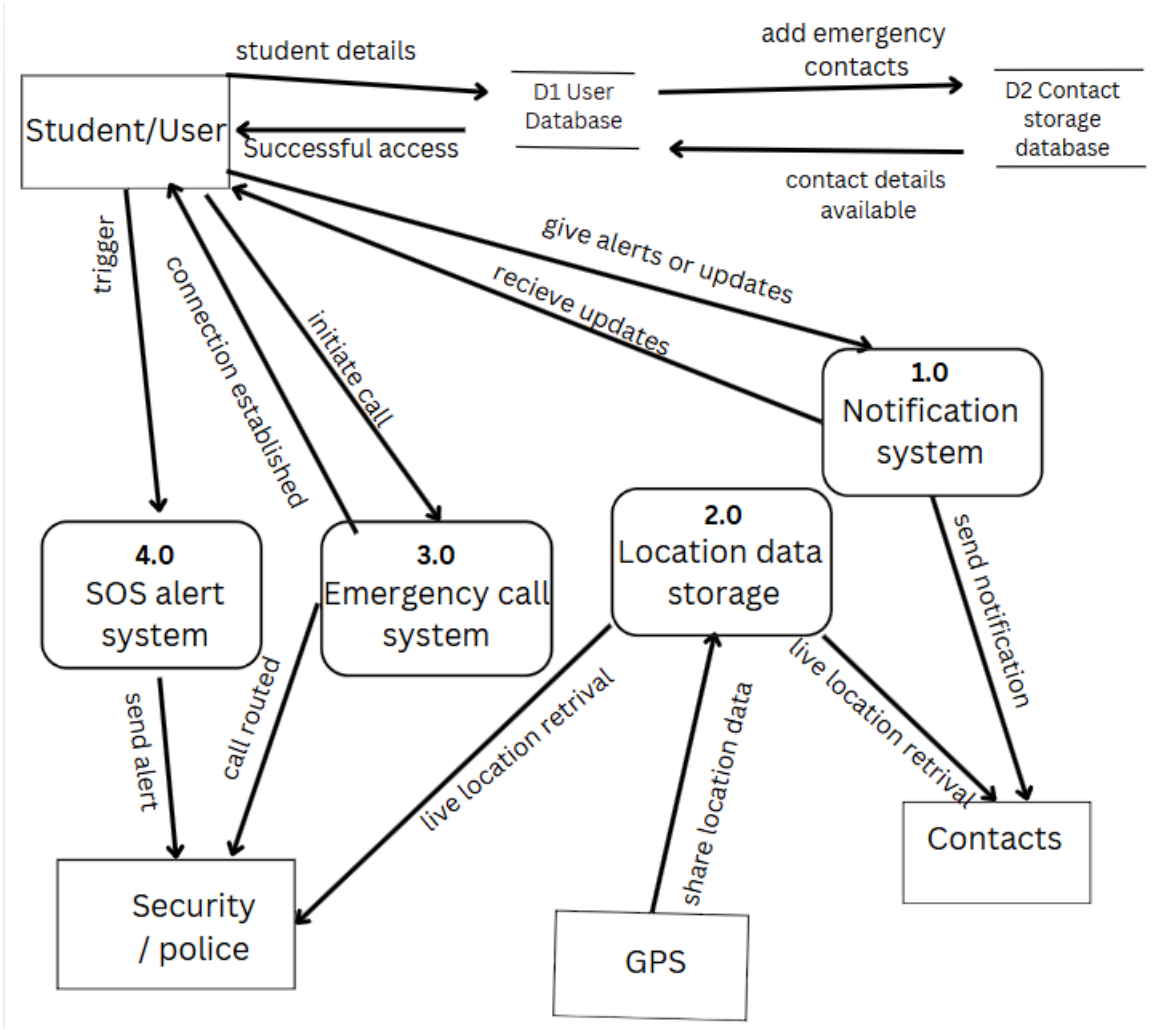
- The context model has Student/User, GPS, Contacts and Security/Police as external entities.
- UoM Safety system as process.
- Dataflows are Registration, Make emergency call, provide contacts access, send SOS alert, get Notifications and share location.



Dataflow diagram:

- The zero level model has Student/User, GPS, Contacts and Security/Police as external entities.
- Location data storage, notification system, emergency call system and SOS alert system are process.
- User database, Contact storage database are datastores.

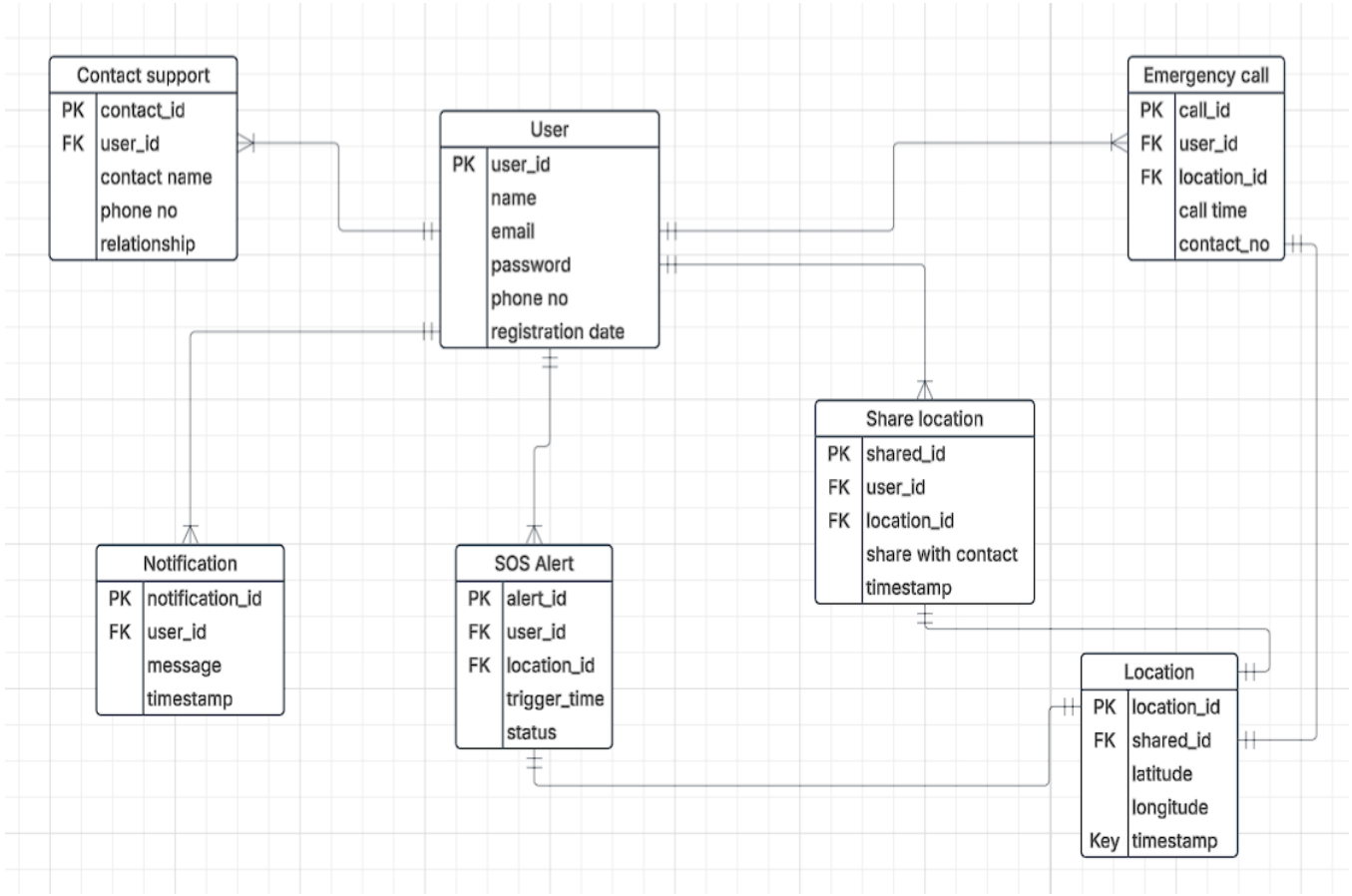
Design Phase



Design Phase

ER diagram:

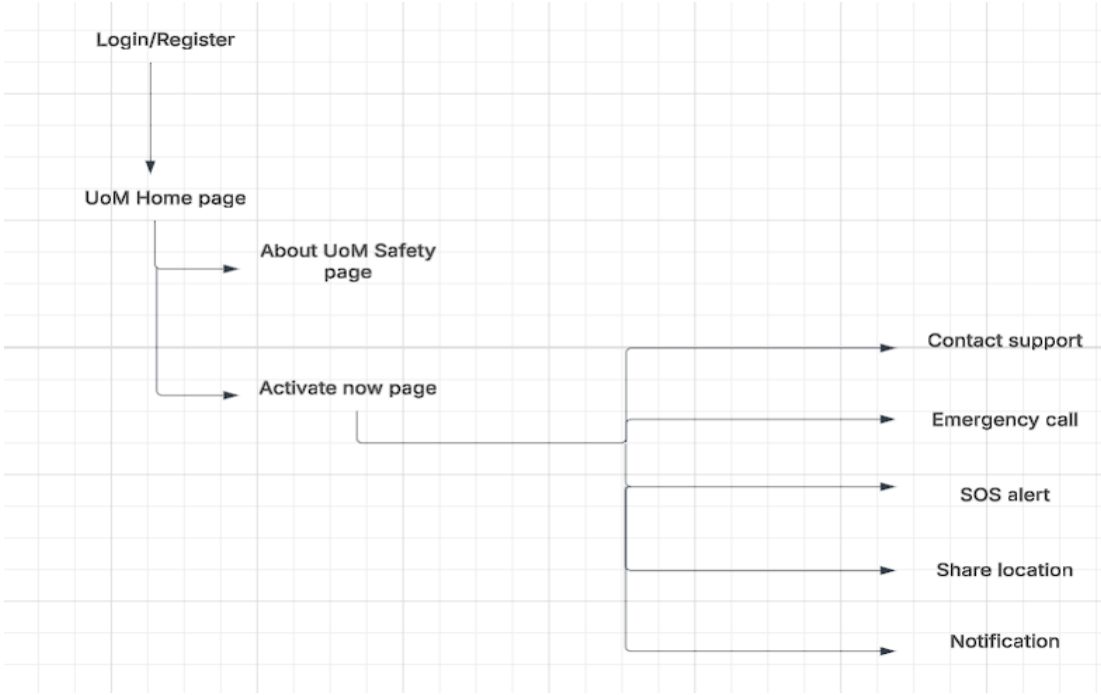
- The User, Contact support, SOS alert, Emergency call, Share location and Notification are all entities and Location is associative entity for ternary relationship among Share location, Emergency call and SOS alert.



Design Phase

Interface navigation map:

- The Flow in user interface design starts from registration and then login, move to UoM home page which consists of UoM about page and Activate now page on click takes user to these pages and the activation page have buttons for making contact support, emergency call, sos alert, share location and notification,



Implementation Phase

Acquisition strategy:

To make the website work effectively, efficiently and smoothly with less investment we have developed inhouse and iterative method with proper testing of frontend and backend and have to host it in cloud.

Alternative matrix:

Matrix 1: Developing inhouse which costs less depends on developer, strong security and customized control with less time to implementation.

Matrix 2: Custom mobile application costs high depending on application developer, web developer and store, strong security, customized control but takes time to implement.

Hence we go with matrix 1 to meet our requirements and budget.

Implementation Phase

Functionality:

- Registering and login is securely made so only the user can access and it also prevent spam attack.
- Contact Support: Saving some contacts of people which later been used to inform student’s or user’s reliable supportive people during emergency.
- Emergency Call: to call nearby safety campus office.
- SOS: Used to send alert immediately for security team.
- Share location: location is by default sent during making emergency call or sos, but if user wants to share location to contacts it will be made separately and they can even view their location.
- Notification: used to get notified from other users who try to contact them.

Implementation Phase

Database storage:

- User table holds registration and login page data where authentication and access is handled with id and password.
- In Activate now page, support request table will have contact support data also generates notification to save, emergency call will login and make call directly to safety office providing location sharing when the call is made and for SOS alert there will be trigger with location sharing.
- Share location logs into with location timestamps and notification is given from notification table.
- In UoM home page the information is displayed so data storage is not required and for UoM safety page contacts links are provided which can be managed dynamically.

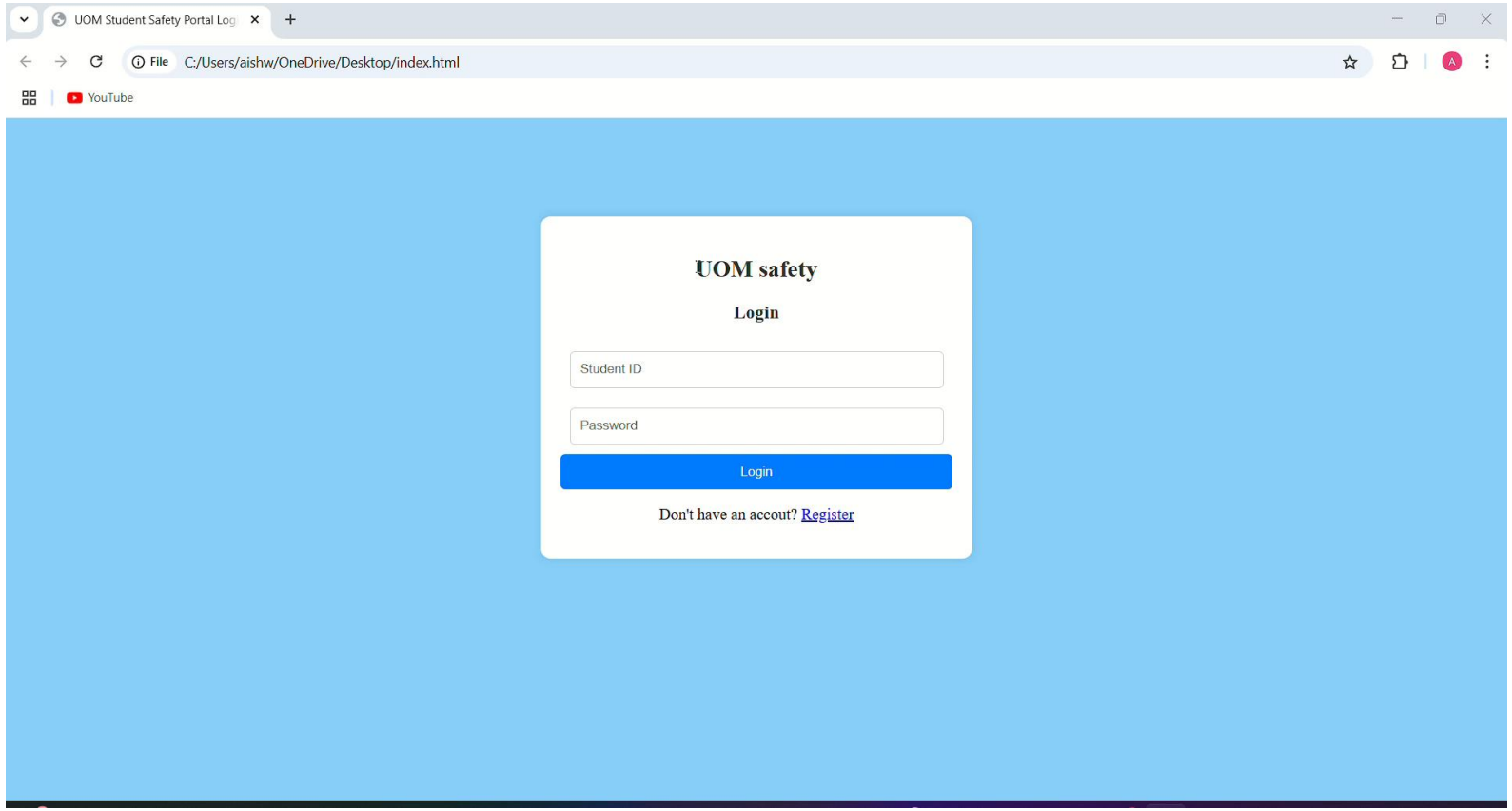
Implementation Phase

System testing: We have implemented black box testing for our proposed system testing the core features and identifying scenarios.

Test cases	Input	Output expected	Test result
1. New user registration	Enter valid information id, name, mail and provide password	Account successfully created then direct to login	Pass
2. Login	Enter registered id and password	Login successfully then go to home page	Pass
3. Provide support contact	Ask permission to link with contacts available in user end	Able to access contacts then make call if needed.	Pass
4. Share Location	Press share	Able to share location and show location	Pass

Testcases	Input	Output expected	Test result
5. Emergency call	Make emergency call	Call the university police and connect immediately	Pass
6. Notification	Press notification	Get notification from any other user in surrounding who needs help, else show no new notification	Pass
7. SOS alert	Press SOS	Alert message sent to UoM Security	Pass

Prototype Video



Conclusion

The project proposes novel approach for our problem, UoM Student safety project is able to replace other old tradition methods followed by the university which were effective but sometimes not much efficient. It have features like Contact support, Emergency call, SOS, Share location and notification. This portal helps the students and also the faculty/ staff to be aware of happenings in and around the campus and also protect the individual providing easy and quick identification of incident and also helps university police/administration take action early as possible.

Thankyou

University of Memphis
Dr. Yafang Li
Jyothi Mallika Golla

And Our dear viewers